

Exercises on Binary Numbers and Boolean Logic

Introduction to Computer Programming

Management and Artificial Intelligence



Recap: binary numbers

Number Systems: Positional vs. Non-Positional

- **Non-Positional Systems:** The value of a symbol is independent of its position.
 - Example: In the Roman numeral **XXVIII** (28)
 - both **X** symbols represent 10
- **Positional Systems:** The value of a symbol depends on its position within the number.
 - Example: In the decimal number **37579** :
 - the first **7** represents 7000
 - while the second **7** represents 70

General Positional System

- Any natural number $b > 1$ can be used as the **base** of a positional number system.
- A number is written as a sequence of digits $d_{n-1}d_{n-2}\dots d_1d_0$, where each digit d_i is an integer from 0 to $b - 1$.
- The value of the number is calculated using the formula:

$$N = \sum_{i=0}^{n-1} d_i \times b^i = d_{n-1}b^{n-1} + d_{n-2}b^{n-2} + \dots + d_1b^1 + d_0b^0$$

The Binary System (Base 2)

When the base $b = 2$, we get the **binary system**. The only possible digits are 0 and 1.

Example: Convert the binary number 11011_2 to decimal

$$\begin{aligned}(11011)_2 &= 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 \\ &= 16 + 8 + 0 + 2 + 1 = (27)_{10}\end{aligned}$$

Bits and Bytes

- **Bit**: A single binary digit (0 or 1).
- **Byte**: A group of 8 bits.

With one byte, we can represent $2^8 = 256$ different values (from 0 to 255). This is enough to represent all the characters on a standard keyboard using an encoding like ASCII.

Octal and Hexadecimal Systems

These systems are commonly used in computing as a compact way to represent binary numbers.

- **Octal:** Uses $b = 8$ and digits 0-7.
- **Hexadecimal:** Uses $b = 16$ and digits 0-9 and letters A-F, where:
 - A = 10, B = 11, C = 12, D = 13, E = 14, F = 15

Converting from Decimal to Another Base

To convert a decimal number N_{10} to base b , we use the method of **successive divisions**:

1. Divide N_{10} by the base b . The remainder is the rightmost digit.
2. Replace N_{10} with the quotient from the division.
3. Repeat until the quotient is 0. The remainders, read in reverse order, form the number in base b .

Example: Convert 116_{10} to Binary

$$116 \div 2 = 58 \Rightarrow \text{remainder } \mathbf{0}$$

$$58 \div 2 = 29 \Rightarrow \text{remainder } \mathbf{0}$$

$$29 \div 2 = 14 \Rightarrow \text{remainder } \mathbf{1}$$

$$14 \div 2 = 7 \Rightarrow \text{remainder } \mathbf{0}$$

$$7 \div 2 = 3 \Rightarrow \text{remainder } \mathbf{1}$$

$$3 \div 2 = 1 \Rightarrow \text{remainder } \mathbf{1}$$

$$1 \div 2 = 0 \Rightarrow \text{remainder } \mathbf{1}$$

Reading the remainders from bottom to top, we get: $(1110100)_2$.

Example: Convert 1547_{10} to Hexadecimal

$1547 \div 16 = 96 \Rightarrow$ remainder **11** (which is **B** in hex)

$96 \div 16 = 6 \Rightarrow$ remainder **0**

$6 \div 16 = 0 \Rightarrow$ remainder **6**

Reading the remainders from bottom to top, we get: $(60B)_{16}$.

Exercises

Possible exam questions!

to Binary

Convert the decimal number **24** to binary:

24_{10} to Binary

Convert the decimal number **24** to binary:

$$24 \div 2 = 12 \text{ (rem 0)}$$

24_{10} to Binary

Convert the decimal number **24** to binary:

$$24 \div 2 = 12 \text{ (rem 0)}$$

$$12 \div 2 = 6 \text{ (rem 0)}$$

24_{10} to Binary

Convert the decimal number **24** to binary:

$$24 \div 2 = 12 \text{ (rem 0)}$$

$$12 \div 2 = 6 \text{ (rem 0)}$$

$$6 \div 2 = 3 \text{ (rem 0)}$$

24_{10} to Binary

Convert the decimal number **24** to binary:

$$24 \div 2 = 12 \text{ (rem 0)}$$

$$12 \div 2 = 6 \text{ (rem 0)}$$

$$6 \div 2 = 3 \text{ (rem 0)}$$

$$3 \div 2 = 1 \text{ (rem 1)}$$

24_{10} to Binary

Convert the decimal number **24** to binary:

$$24 \div 2 = 12 \text{ (rem 0)}$$

$$12 \div 2 = 6 \text{ (rem 0)}$$

$$6 \div 2 = 3 \text{ (rem 0)}$$

$$3 \div 2 = 1 \text{ (rem 1)}$$

$$1 \div 2 = 0 \text{ (rem 1)}$$

24_{10} to Binary

Convert the decimal number **24** to binary:

$$24 \div 2 = 12 \text{ (rem 0)}$$

$$12 \div 2 = 6 \text{ (rem 0)}$$

$$6 \div 2 = 3 \text{ (rem 0)}$$

$$3 \div 2 = 1 \text{ (rem 1)}$$

$$1 \div 2 = 0 \text{ (rem 1)}$$

Result: $(11000)_2$

to Binary

To convert from decimal to binary:

63_{10} to Binary

To convert 63 from decimal to binary:

$$63 \div 2 = 31 \text{ remainder } 1$$

$$31 \div 2 = 15 \text{ remainder } 1$$

$$15 \div 2 = 7 \text{ remainder } 1$$

$$7 \div 2 = 3 \text{ remainder } 1$$

$$3 \div 2 = 1 \text{ remainder } 1$$

$$1 \div 2 = 0 \text{ remainder } 1$$

Reading from bottom to top: 111111

Therefore, $63_{10} = 111111_2$

to Binary

To convert from decimal to binary:

129_{10} to Binary

To convert 129 from decimal to binary:

$$129 \div 2 = 64 \text{ remainder } 1$$

$$64 \div 2 = 32 \text{ remainder } 0$$

$$32 \div 2 = 16 \text{ remainder } 0$$

$$16 \div 2 = 8 \text{ remainder } 0$$

$$8 \div 2 = 4 \text{ remainder } 0$$

$$4 \div 2 = 2 \text{ remainder } 0$$

$$2 \div 2 = 1 \text{ remainder } 0$$

$$1 \div 2 = 0 \text{ remainder } 1$$

Reading from bottom to top: 10000001

Therefore, $129_{10} = 10000001_2$

to Decimal

Convert the binary number **10101** to decimal:

10101_2 to Decimal

Convert the binary number **10101** to decimal:

$$(10101)_2 = 1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0$$

$$= 16 + 0 + 4 + 0 + 1 = (21)_{10}$$

to Decimal

Convert the binary number **100100** to decimal:

100100_2 to Decimal

Convert the binary number **100100** to decimal:

$$(100100)_2 = 1 \cdot 2^5 + 0 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 0 \cdot 2^0$$

$$= 32 + 0 + 0 + 4 + 0 + 0 = (36)_{10}$$

to Decimal

Convert the binary number **11100111** to decimal:

1100111_2 to Decimal

Convert the binary number **11100111** to decimal:

$$(11100111)_2 = 1 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^5 + 0 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$$

$$= 128 + 64 + 32 + 0 + 0 + 4 + 2 + 1 = (231)_{10}$$

to Decimal

Convert the hexadecimal number **1547** to decimal:

1547_{16} to Decimal

Convert the hexadecimal number **1547** to decimal:

$$(1547)_{16} = 1 \cdot 16^3 + 5 \cdot 16^2 + 4 \cdot 16^1 + 7 \cdot 16^0$$

$$= 1 \cdot 4096 + 5 \cdot 256 + 4 \cdot 16 + 7 \cdot 1$$

$$= 4096 + 1280 + 64 + 7 = (5447)_{10}$$

to Decimal

Convert the hexadecimal number **CAFE** to decimal:

$CAFE_{16}$ to Decimal

Convert the hexadecimal number **CAFE** to decimal:

$$(CAFE)_{16} = C \cdot 16^3 + A \cdot 16^2 + F \cdot 16^1 + E \cdot 16^0$$

$CAFE_{16}$ to Decimal

Convert the hexadecimal number **CAFE** to decimal:

$$(CAFE)_{16} = C \cdot 16^3 + A \cdot 16^2 + F \cdot 16^1 + E \cdot 16^0$$

$$= 12 \cdot 4096 + 10 \cdot 256 + 15 \cdot 16 + 14 \cdot 1$$

$CAFE_{16}$ to Decimal

Convert the hexadecimal number **CAFE** to decimal:

$$(CAFE)_{16} = C \cdot 16^3 + A \cdot 16^2 + F \cdot 16^1 + E \cdot 16^0$$

$$= 12 \cdot 4096 + 10 \cdot 256 + 15 \cdot 16 + 14 \cdot 1$$

$$= 49152 + 2560 + 240 + 14 = (51966)_{10}$$

*DEADBEEF*₁₆ to Decimal

Convert the hexadecimal number **DEADBEEF** to decimal:

$$\begin{aligned} DEADBEEF_{16} &= D \cdot 16^7 + E \cdot 16^6 + A \cdot 16^5 + D \cdot 16^4 + B \cdot 16^3 + E \cdot 16^2 + E \cdot 16^1 + F \\ &\quad \cdot 16^0 \\ &= 13 \cdot 16^7 + 14 \cdot 16^6 + 10 \cdot 16^5 + 13 \cdot 16^4 + 11 \cdot 16^3 + 14 \cdot 16^2 + 14 \cdot 16^1 + 15 \cdot 16^0 \\ &= 13 \cdot 268435456 + 14 \cdot 16777216 + 10 \cdot 1048576 + 13 \cdot 65536 + 11 \cdot 4096 + 14 \cdot 256 + 14 \cdot \\ &\quad + 15 \cdot 1 \\ &= 3489660928 + 234881024 + 10485760 + 851968 + 45056 + 3584 + 224 + 15 \\ &= 3735928559_{10} \end{aligned}$$

Recap: Boolean Logic

Boolean Operators and Truth Tables

Boolean algebra operates on truth values (True/1, False/0) using logical operators.

- **AND (Conjunction, $\wedge$):** True only if both operands are true.
- **OR (Disjunction, $\vee$):** True if at least one operand is true.
- **NOT (Negation, $\neg$):** Inverts the truth value.

Truth Table Example

Let's evaluate the expression:

Truth Table Example

Let's evaluate the expression: $F = (A \wedge B) \vee (\neg A \wedge C)$

A	B	C	A AND B	!A	!A AND C	F
0	0	0	0	1	0	0
0	0	1	0	1	1	1
0	1	0	0	1	0	0
0	1	1	0	1	1	1
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	1	0	0	1
1	1	1	1	0	0	1

Key Concepts in Boolean Logic

- **Equivalence:** Two expressions are equivalent if they have the same truth table.
 - *Example (De Morgan's Law):* $\neg(A \vee B) \equiv \neg A \wedge \neg B$
- **Tautology:** An expression that is always true, regardless of the input values.
 - *Example:* $A \vee \neg A$
- **Contradiction:** An expression that is always false.
 - *Example:* $A \wedge \neg A$

Exercises

Exercise: Evaluate the expression $\neg A \wedge \neg B \vee (\neg B \wedge A)$

Let's use $A'B' + B'A$ for simplicity.

A	B	A'	B'	A'B'	B'A	A'B' + B'A
0	0	1	1	1	0	1
0	1	1	0	0	0	0
1	0	0	1	0	1	1
1	1	0	0	0	0	0

